

Marko Velimirović (22)

Backend / Full-Stack Engineer

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Portfolio: velimirovic.github.io

SUMMARY

Backend-leaning full-stack engineer with a B.Sc. in Electrical & Computer Engineering from the Faculty of Technical Sciences, Novi Sad. Builds distributed systems in C#, Java, Go, Node.js and Python - the thesis project runs as five containerised microservices in three languages behind one API gateway. Looking for a first engineering role; available from 09/2026.

TECHNICAL SKILLS

Languages: C#, Java, Go, Python, TypeScript, C++, SQL

Backend: ASP.NET Core 9, Spring Boot, Node.js / Express, FastAPI, EF Core, REST, gRPC, SignalR, JWT

Frontend: Angular 16-21, React 18/19, RxJS, SCSS, PWA | **Databases:** PostgreSQL, MongoDB, Neo4j, Redis

Infrastructure: Docker, Docker Compose, RabbitMQ, YARP gateway, GitHub Actions, Git

AI integration: OpenAI API (GPT-4o mini), Claude API, prompt design, output validation, caching and rate limiting

AI tools: **Claude Code** (daily), GitHub Copilot in VS Code, Cursor, Perplexity

Practices: Microservices, modular monolith, Clean Architecture, MVC/MVVM, xUnit, Jest, pytest, Prometheus, Grafana

PROJECTS

Shelvio - Microservice platform with AI recommendations · B.Sc. thesis · 2026

ASP.NET Core 9 · Node.js · Python / FastAPI · Angular 21 · PostgreSQL · Redis · RabbitMQ · Docker Compose · GPT-4o mini · github.com/velimirovic/shelvio

- Built a media tracking and recommendation platform as 5 independently deployable services in 3 languages, behind one ASP.NET Core / YARP gateway handling routing and JWT validation centrally.
- Designed the recommendation pipeline - library grouped into rating tiers, prompted to GPT-4o mini, output filtered against titles already owned - and held model spend down with a 12-hour Redis cache and a per-user daily cap.
- Gave each service its own store across 2 PostgreSQL databases, Redis and RabbitMQ; wrote tests in 3 toolchains; the whole stack starts with one `docker compose up`.

TourFlo - Seven-service microservice platform · team project · 2025

Go · C# · Java / Spring Boot · Node.js · Angular 18 · gRPC · PostgreSQL · MongoDB · Neo4j · Docker Compose · github.com/velimirovic/soa-tourism-app

- Built 7 independently deployable services across 5 runtimes, communicating over REST and gRPC through a Go API gateway that validates JWTs and routes traffic.
- Matched each store to its data - Neo4j for the social graph, MongoDB for the blog, PostgreSQL for tours and accounts - all containerised with Docker Compose.

Travel Vista - Tour platform with real-time tracking · team project · 2025

ASP.NET Core 8 · Angular 16 · PostgreSQL · SignalR · Leaflet · OpenAI API · github.com/kzi-nastava/psw-be-ra-2025-group-3

- Implemented live GPS tracking over SignalR, streaming participant positions onto Leaflet maps during a tour. Built as a modular monolith on Clean Architecture with enforced module boundaries, working across the backend and the frontend in a team.

Nine further projects - a geo-tagged video platform with an ETL pipeline and Grafana dashboards, a conference PWA with a Claude API assistant, a marketplace with live auctions, a WPF desktop app, two Unity games and two OpenGL renderers in C++ - at velimirovic.github.io.

EDUCATION

Faculty of Technical Sciences, University of Novi Sad - B.Sc. Electrical & Computer Engineering

Computing and Automation · 10/2022 - 09/2026 · Thesis: Shelvio (mentor dr Miroslav Zarić)

Electrical Engineering School "Mihajlo Pupin", Novi Sad - IT Technician · 09/2018 - 06/2022

LANGUAGES & ADDITIONAL

Serbian - native · English - professional working proficiency · Member of Mensa